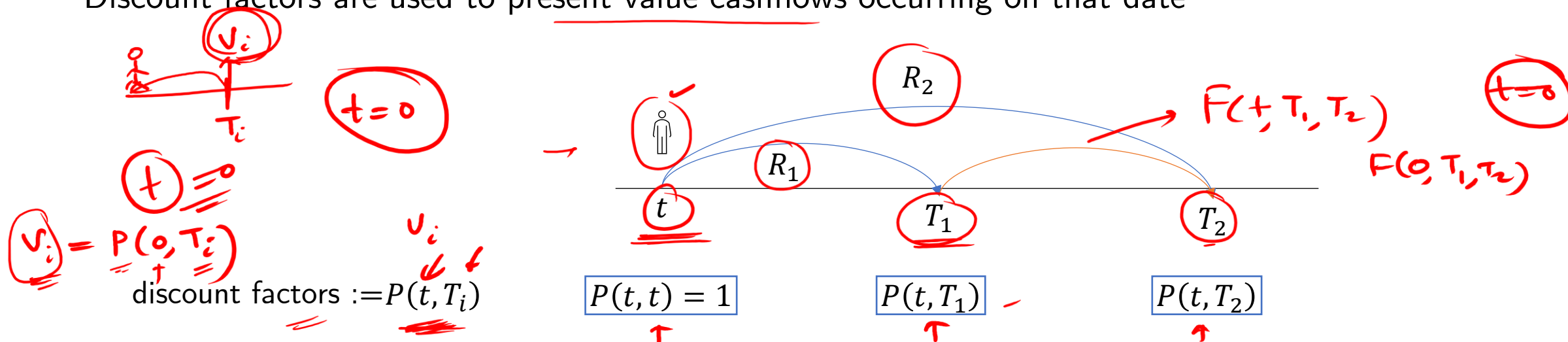


RFR Regime

(4. Single Currency RFR Curves)

Rate preliminaries

- For any date, we can assign a discount factor which is the value of a ZCB maturing that date. Discount factors are used to present value cashflows occurring on that date



- Let R_1 and R_2 be the continuously compounded spot rates for maturities T_1 and T_2 respectively. Then we can obtain continuously compounded forward rate between T_1 and T_2 as

$$\begin{aligned}
 v_1 &= e^{-R_1 T_1} \rightarrow P(t, T_1) = e^{-R_1(T_1 - t)} \\
 v_2 &= e^{-R_2 T_2} \rightarrow P(t, T_2) = e^{-R_2(T_2 - t)}
 \end{aligned}
 \left. \vphantom{\begin{aligned} v_1 &= e^{-R_1 T_1} \\ v_2 &= e^{-R_2 T_2} \end{aligned}} \right\} \rightarrow \underline{\underline{F(t, T_1, T_2)}} = \frac{R_2(T_2 - t) - R_1(T_1 - t)}{T_2 - T_1} = - \frac{\ln P(t, T_2) - \ln P(t, T_1)}{T_2 - T_1}$$

Rate preliminaries

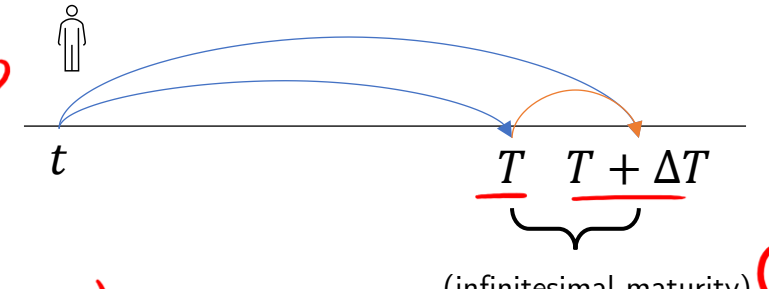
- We can define instantaneous forward rate as the forward rate applicable for an infinitesimal maturity. The instantaneous forward rate is given as

Handwritten notes and diagrams illustrating the instantaneous forward rate:

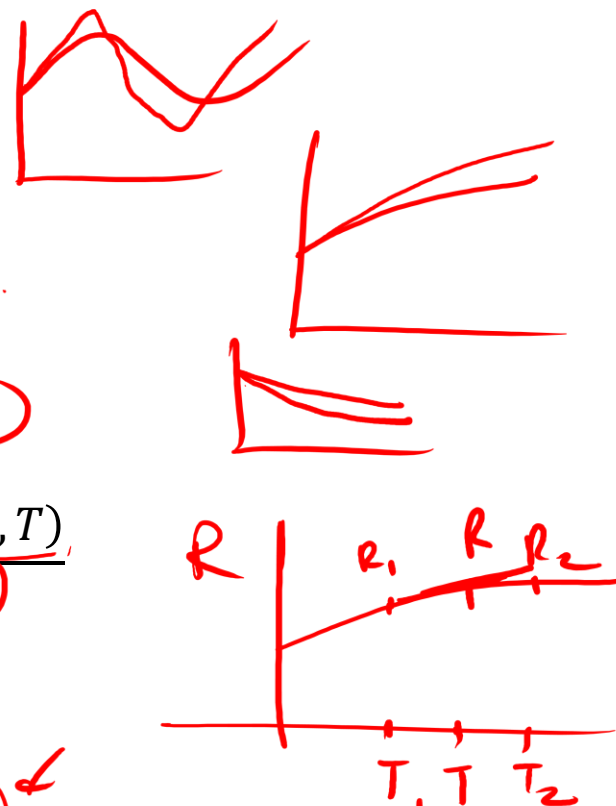
Left side notes:

- $\prod_{i=1}^n (1 + \delta_i r_i)$
- $f(x) = \frac{f(x+\Delta x) - f(x)}{(x+\Delta x) - x}$
- $F(t, T, T+\Delta T)$
- $f(t, T)$

Diagram:



Right side diagrams:



Equation for instantaneous forward rate:

$$f(t, T) = - \frac{\ln P(t, T + \Delta T) - \ln P(t, T)}{T + \Delta T - T} = - \frac{\partial \ln P(t, T)}{\partial T}$$

Equation for forward rate in terms of spot rate:

$$f(t, T) = \frac{\partial (R(T - t))}{\partial T} = R + (T - t) \frac{dR}{dT}$$

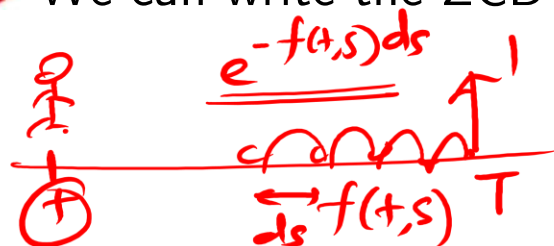
Left side equations for spot rate:

$$P(t, T) = e^{-R(T-t)}$$

$$-\ln P = R(T-t)$$

Rate preliminaries

- We can write the ZCB price or the spot rate as integrals of the instantaneous forward rate.



$$P(t, T) = \exp\left(-\int_t^T f(t, s) ds\right)$$

$$f(t, T) = -\frac{\partial \ln P(t, T)}{\partial T}$$

Which immediately tells us that the spot rate is the average of instantaneous forward rates

$$P(t, T) = e^{-R(t, T)(T-t)}$$

$$R = \frac{1}{T-t} \int_t^T f(t, s) ds$$

$$R(t, T) = \frac{1}{T-t} \int_t^T f(t, s) ds$$

Also note that



$$R_2(T_2 - t) = R_1(T_1 - t) + \int_{T_1}^{T_2} f(t, s) ds$$

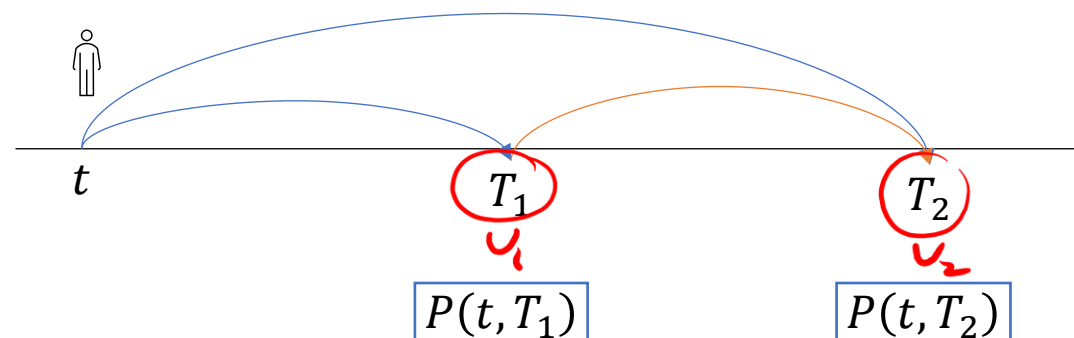
Rate preliminaries

- The simply compounded forward rate is used to calculate floating coupon payments. It's conveniently given using discount factors as

Diagram showing a timeline from 0 to T_2 with a rate R indicated above the timeline.

$$e^{(R)T_2} = \left(1 + \frac{R' \times T_2}{1}\right)$$

cond. simply



$$F(0, T_1, T_2) = \frac{1}{\tau(T_1, T_2)} \left(\frac{v_1}{v_2} - 1 \right)$$

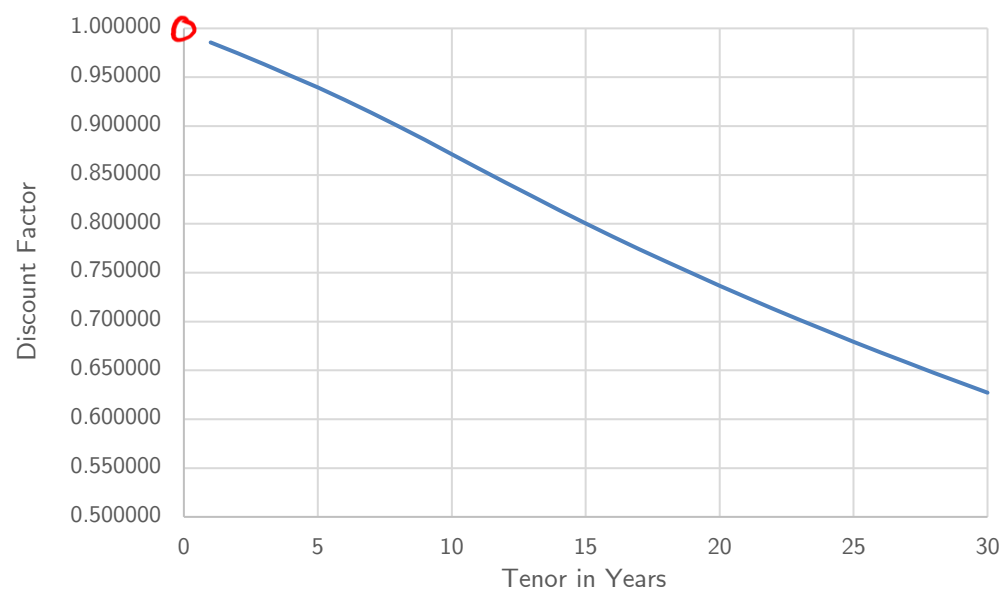
$$\underline{\underline{F(t, T_1, T_2)}} = \frac{1}{\underline{\underline{\tau(T_1, T_2)}}} \left(\frac{P(t, T_1)}{P(t, T_2)} - 1 \right)$$

Curve Modelling

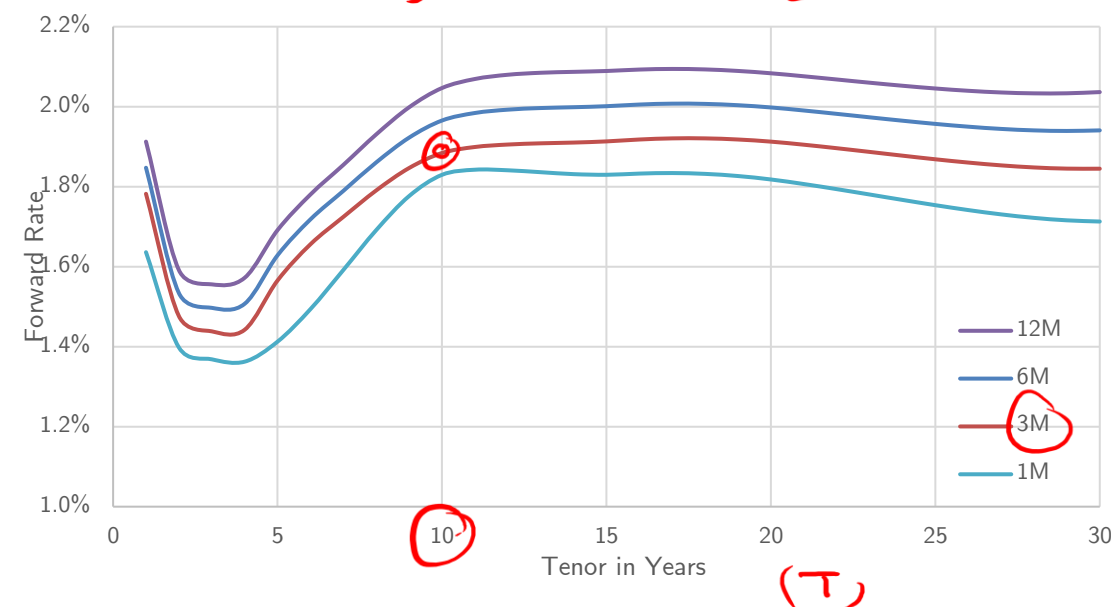
- ▶ Banks maintain many curves such as zero-curve (spot rate curve), forward curves, par rate curves, basis curves etc. The zero-curve is used to discount the cashflows. The forward curve is used to project cashflows of the floating rate payments. In the RFR regime, single curve per currency is becoming common, so we will focus on a single RFR curve per currency.
- ▶ Discount factors can be defined for each date. Forward rate curves can be easily derived from the discount curve for a specific maturity starting at different times in future.

V
 T

RFR discount curve



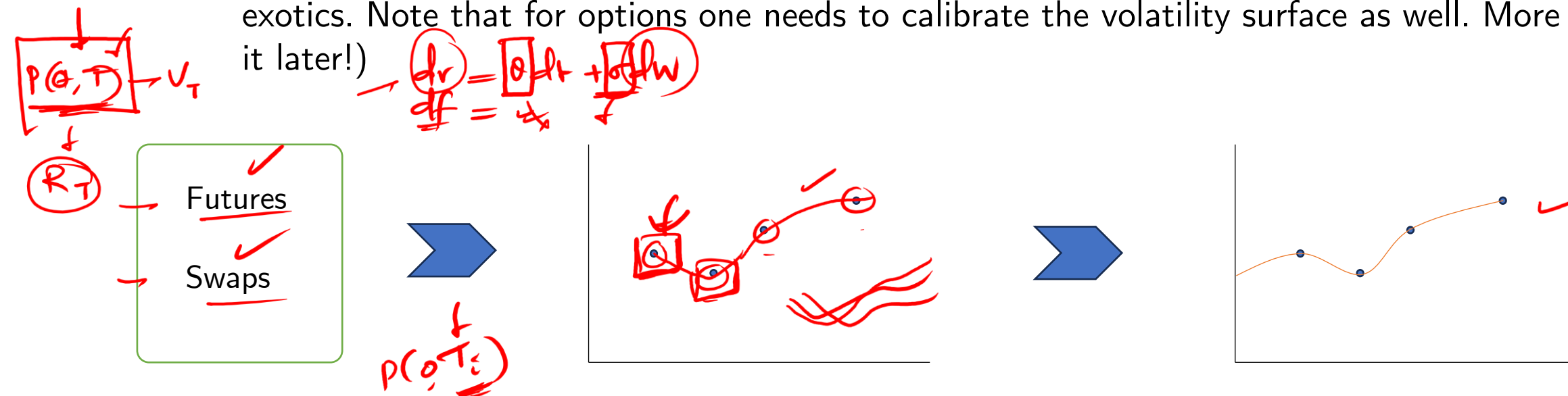
$F(0, 10, 10.25)$ RFR Forward Rates



$F(t, T, T+\Delta)$

Curve Modelling

- ▶ For building a single currency curve, one typically follows the steps below
 - ▶ Identify liquid instruments (e.g. SOFR futures for the short end, swaps for long end)
 - ▶ Bootstrap zero rates for the maturities applicable to the instruments above.
 - ▶ To create a continuous curve, there are 2 broad approaches
 - ▶ Non-Model (devise an interpolation scheme to fit the discrete points)
 - ▶ Model (devise a stochastic process for the rate (term structure models) such that the price of ZCBs from the model perfectly match the market quotes. Then use the model to price exotics. Note that for options one needs to calibrate the volatility surface as well. More on it later!)



Curve Modelling Choices

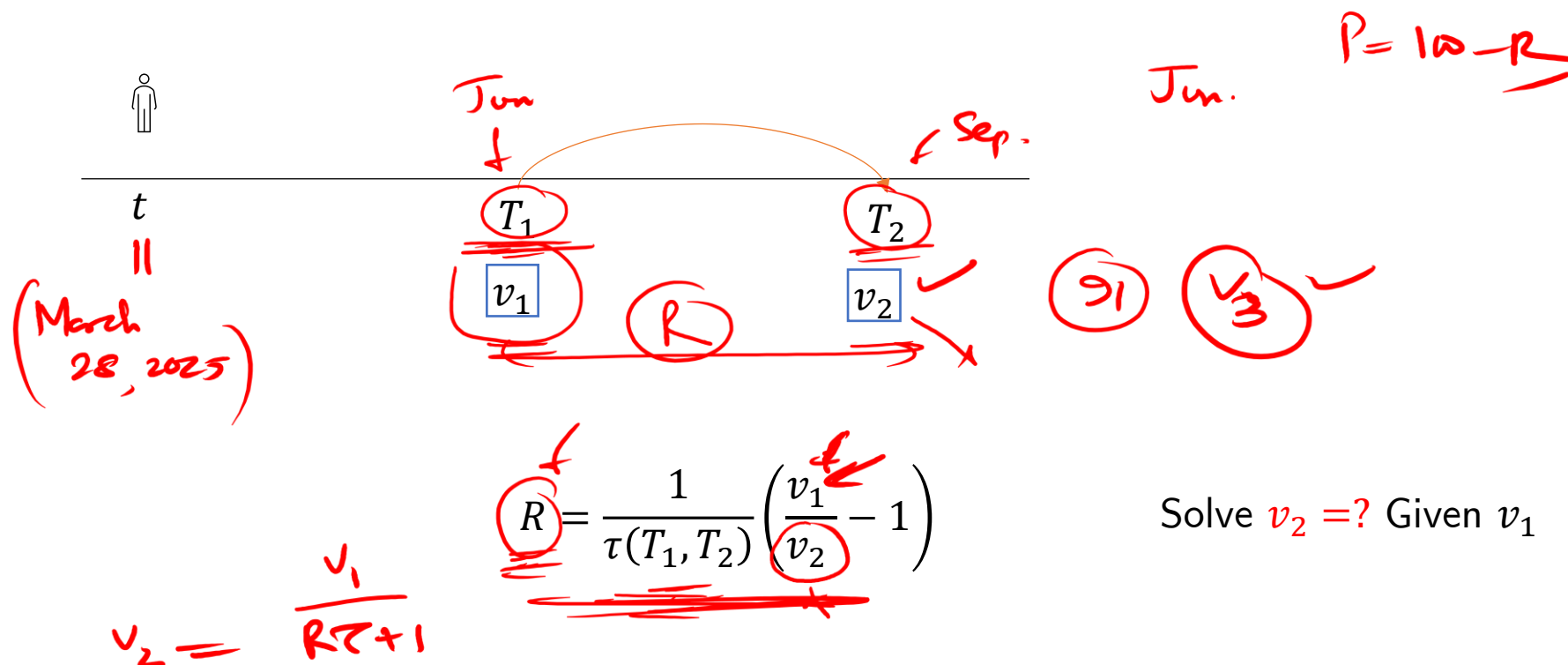
- ▶ In this exercise, we will focus on single currency bootstrap procedure and then a suitable interpolation scheme to generate a continuous curve. There are 2 main considerations

- ▶ Base: zero rates, forward rates or discount factor
- ▶ Interpolation:
 - ▶ Linear ✓
 - ▶ Log-Linear ✓
 - ▶ Splines (Natural Cubic spline, B-spline, Hermite spline)
 - ▶ Monotone Convex

→ Pat Hagan → SABR

Bootstrap DFs from SOFR futures - I

- SOFR 3M futures pack the information of the forward rate between the IMM dates of contract month and delivery month. First, we find the forward rate $R = 100 - P$. Then we bootstrap the discounting factor recursively which apply to the start and end of the above period.



Bootstrap DFs from SOFR futures - II

- ▶ If the calibration date falls between the reference quarter of a future, then we need to take care of the SOFR fixings that have already taken prior to the calibration date.
- ▶ Obtain the total compounding factor for the published SOFR rates. Use it in conjunction with expected forward rate for the remaining period to obtain the discount factor.

Handwritten notes and diagram illustrating the bootstrap process for discount factors (DFs) from SOFR futures.

Diagram: A timeline showing dates: March 19th (T_1), April 5th (t), and June 17th (T_2). A person icon is at t . Green 'x' marks represent SOFR fixings before t . A red arrow points from T_2 to t , labeled v_2 (March). A blue box labeled v_2 is at T_2 . A red arrow points from T_2 to t , labeled R (12-P).

Equations:

$$R = \left[\prod_{i=1}^n (1 + d_i r_i) - 1 \right] \times \frac{360}{n_c}$$

$$\left(\prod_{i=1}^{n_1} (1 + d_i r_i^*) \left(\frac{1}{v_2} - 1 \right) \right) \times \left(\frac{360}{d} \right) = R$$

Solve $v_2 = ?$

$$\left(1 + \frac{Rd}{360} \right) = \frac{\prod (1 + d_i r_i^*)}{v_2}$$

$$v_2 = \frac{\prod (1 + d_i r_i^*)}{1 + \frac{Rd}{360}}$$

FRB NY
CME

Bootstrap DFs from SOFR Swaps

- ▶ We will choose the Annual/Annual SOFR swaps which are most liquid. Swap rates for liquid tenors (1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 15Y, 30Y) can be used to bootstrap the discount factors using the closed form formula for swap rates

$$s_n = \frac{1 - v_n}{\tau \sum_{i=1}^n v_i}$$

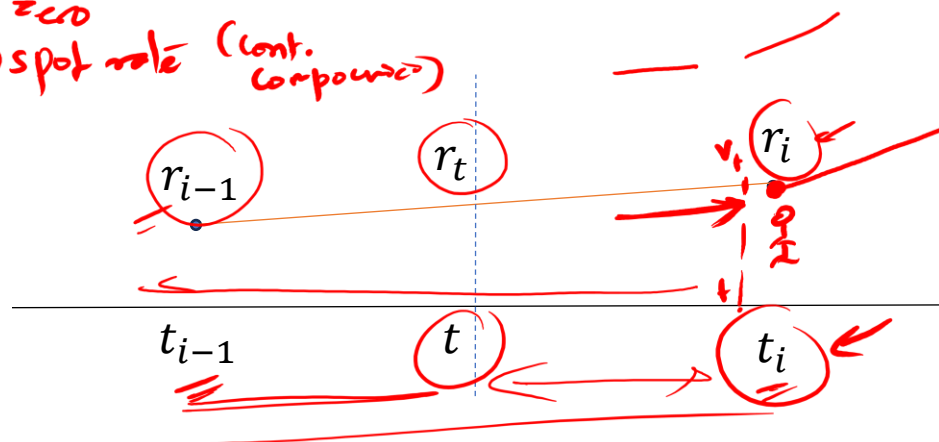
Exercise – Show that v_n satisfies the recursive formula

$$v_n = \frac{1 - \tau s_n \sum_{i=1}^{n-1} v_i}{1 + \tau s_n}$$

Linear Methods (Rates)

- For $t_{i-1} \leq t \leq t_i$, the interpolation formula would be

$r \rightarrow$ ^{zero} spot rate (cont. compounding)



$$r_t = \frac{t - t_{i-1}}{t_i - t_{i-1}} r_i + \frac{t_i - t}{t_i - t_{i-1}} r_{i-1}$$

$$f_t = \frac{2t - t_{i-1}}{t_i - t_{i-1}} r_i + \frac{t_i - 2t}{t_i - t_{i-1}} r_{i-1}$$

- The instantaneous forward f is undefined (has jumps) at t_i because rt is not differentiable.
- This does not stop forward rates from becoming negative.

$$\frac{\partial (r_t \cdot t)}{\partial t} = f(t, t)$$



$v_i \rightarrow r_i$

v_i

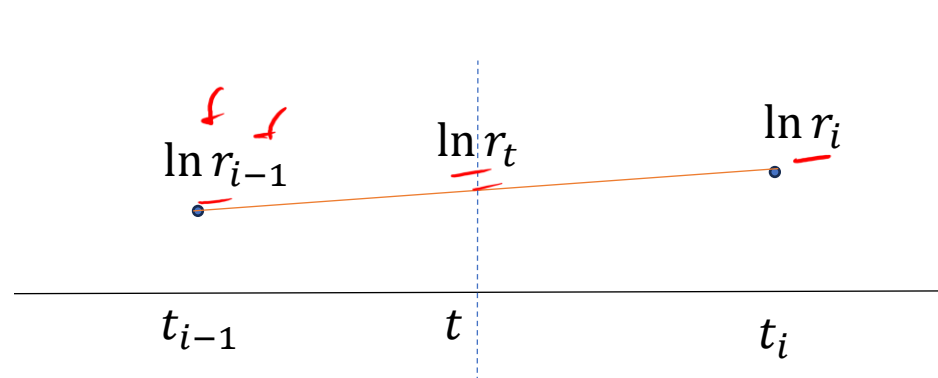
T_i

$e^{-r_i T_i} = v_i$

$r_i = -\frac{1}{T_i} (\ln v_i)$

Linear Methods (Log Rates)

- For $t_{i-1} \leq t \leq t_i$, the interpolation formula would be



$r_i > 0$

$$\ln r_t = \frac{t - t_{i-1}}{t_i - t_{i-1}} \ln r_i + \frac{t_i - t}{t_i - t_{i-1}} \ln r_{i-1}$$

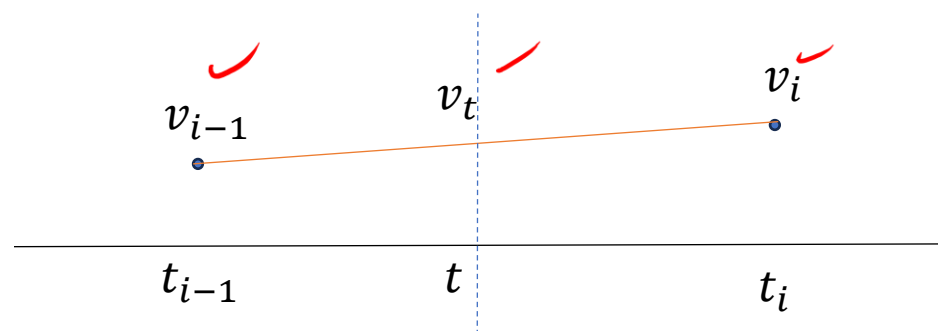
$f_+ = ?$

- The instantaneous forward f is undefined (has jumps)
- Negative spot rates are not allowed
- Discount factor may not be decreasing



Linear Methods (Discount Factors)

- For $t_{i-1} \leq t \leq t_i$, the interpolation formula would be



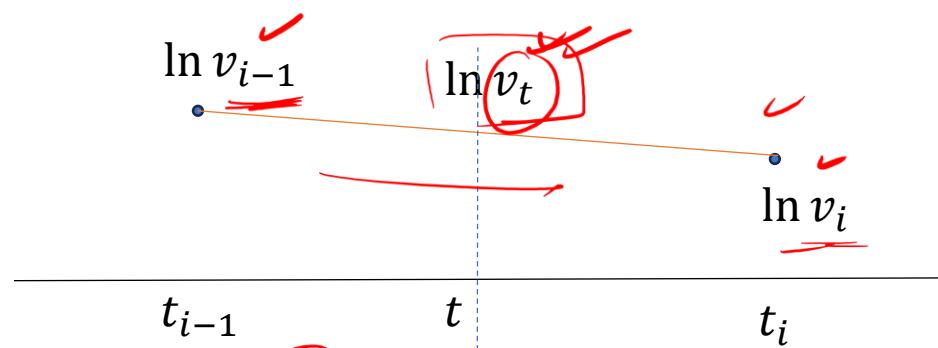
$$\rightarrow \underline{v_t} = \frac{t - t_{i-1}}{t_i - t_{i-1}} v_i + \frac{t_i - t}{t_i - t_{i-1}} v_{i-1}$$

$$\rightarrow r_t = \left(-\frac{1}{t} \right) \ln \left(\frac{t - t_{i-1}}{t_i - t_{i-1}} e^{-r_i t_i} + \frac{t_i - t}{t_i - t_{i-1}} e^{-r_{i-1} t_{i-1}} \right)$$

- The instantaneous forward f is undefined (has jumps) at t_i 
- Also, discount factors (v) may not be decreasing

Linear Methods (Log Discount Factors)

- For $t_{i-1} \leq t \leq t_i$, the interpolation formula would be



$$\ln v_t = \frac{t - t_{i-1}}{t_i - t_{i-1}} \ln v_i + \frac{t_i - t}{t_i - t_{i-1}} \ln v_{i-1}$$

$$r_t t = \frac{t - t_{i-1}}{t_i - t_{i-1}} r_i t_i + \frac{t_i - t}{t_i - t_{i-1}} r_{i-1} t_{i-1}$$

$$f_t = -\frac{\partial \ln v}{\partial t} > 0 \quad v_t = e^{-r_t \cdot t} \quad , \quad \ln v = -r_t \cdot t$$

- The instantaneous forward f is undefined (has jumps) at t_i
- It's constant in every interval and is equal to the discrete forward rate
- Also, it's guaranteed to be positive

$$F(0, T, T + \Delta T) = F(0, T_{i-1}, T_i)$$



$$f_t$$

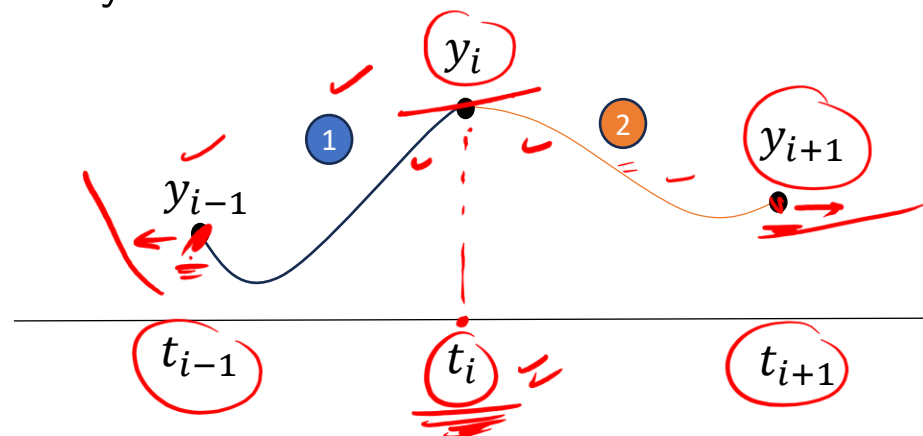
$$\begin{aligned} & \frac{\partial}{\partial t} (r_t \cdot t) \\ &= \frac{r_i t_i}{t_i - t_{i-1}} - \frac{r_{i-1} t_{i-1}}{t_i - t_{i-1}} \\ &= \boxed{\frac{r_i t_i - r_{i-1} t_{i-1}}{t_i - t_{i-1}}} \end{aligned}$$

Splines

- ▶ Cubic splines are universally used as **smooth** interpolation tools.
- ▶ For every interval, we could assign a cubic polynomial. Depending on how smooth interpolation we want we could enforce constraints accordingly. The following definitions are useful

C^0 : Value continuity ✓ ① $a_1x^3 + b_1x^2 + c_1x + d_1$ $- 3a_1x^2 + 2b_1x + c_1$
 C^1 : Slope continuity ✓ ② $a_2x^3 + b_2x^2 + c_2x + d_2$ $6a_2x + 2b_2$
 C^2 : Curvature continuity ✓

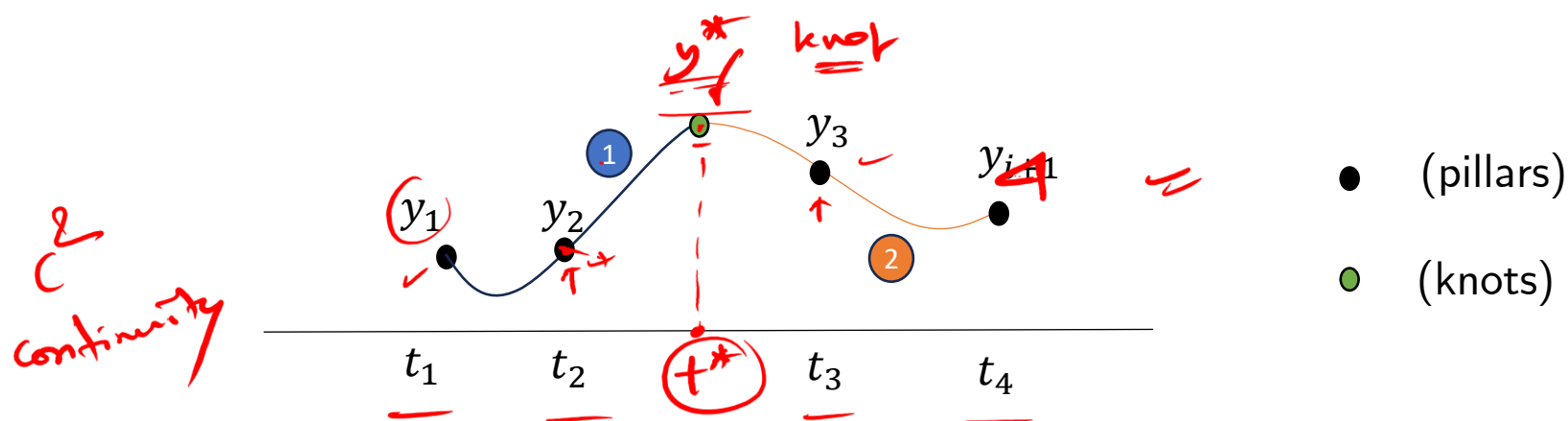
① $a_1x_i^3 + b_1x_i^2 + c_1x_i + d_1$
 $= a_2x_i^3 + b_2x_i^2 + c_2x_i + d_2$
 ② $3a_1x_i^2 + 2b_1x_i + c_1$
 $= 3a_2x_i^2 + 2b_2x_i + c_2$
 ③ $6a_1x_i + 2b_1 = 6a_2x_i + 2b_2$



- ▶ A common cubic spline has C^2 continuity. A Natural cubic spline also ensures that second derivative is 0 at both ends.

Pillars vs Knots

- ▶ It is not necessary that we choose one cubic for each interval between the data points. It can be arbitrary. It's important to distinguish between **pillars** and **knots**.
- ▶ Knots are the points where spline segments meet, while pillar points are the data points that the spline is designed to fit or interpolate. For example, in the graphic below, we have 4 pillar points, but only 1 knot. Knots determine how many polynomials are needed.



Degree of Freedom and Constraints

- ▶ Let's suppose we want to fit a Natural cubic spline to a set of n data points.
- ▶ We have $(n - 1)$ intervals, and we choose $(n - 1)$ cubic polynomials of the form

$$\underline{a_i}x^3 + \underline{b_i}x^2 + \underline{c_i}x + \underline{d_i}$$

$$\underline{\underline{n}}$$

$$\underline{\underline{n-2}}$$

- ▶ Total degree of freedom = $4 \times (n - 1) = \underline{\underline{4n - 4}}$

- ▶ There are also a total of $(4n - 4)$ constraints

- Each curve ^{polynomial} passes through 2 points (start and end of the intervals) = $2 \times (n - 1) = \underline{\underline{2n - 2}}$
- At every knot except first and last, we need C^2 continuity = $2 \times (n - 2) = \underline{\underline{2n - 4}}$
- At both ends, we need the second derivative to be zero = 2

$\underline{\underline{4n - 4}}$ equations, $\underline{\underline{4n - 4}}$ unknowns. Use a simple matrix equation to solve the system of linear equations.